CS 215: Task 02 Due: 15-Aug Time: as mentioned in Piazza. No extensions.

0. Please write the honor code in your handwriting, with the understanding that it is true.

1. X is a continuous random variable with pdf

$$f(x) = \begin{cases} C(4x - 2x^2), & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

(a) C is

C is 3/8.

Teaching Team Notes: It is advisable to answer the question :). This implies that, in this case, you state what is C is (either at the end or at the beginning, but state the answer).

Since f is a probability density function, we must have $\int_{-\infty}^{\infty} f(x)\,dx = 1$, which means

$$C \int_0^2 (4x - 2x^2)\,dx = 1$$

$$C \left[2x^2 - \frac{2x^3}{3}\right]_{x=0}^{x=2} = 1$$

$$C = \frac{3}{8}$$

(b) $\mathbb{P}\{X > 1\}$ is

$$P\{X > 1\} = \int_1^{\infty} f(x)\,dx = \frac{3}{8}\int_1^2 (4x - 2x^2)\,dx = \frac{1}{2}$$

2. The amount of time, in hours, that a computer functions before breaking down is a continuous random variable with pdf

$$f(x) = \begin{cases} \lambda e^{-x/100}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

(a) What is the probability that a computer will function between 50 and 150 hours before breaking down?

Since $f(x)$ is a probability density function,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1.$$

Thus,

$$\lambda \int_0^\infty e^{-x/100}\,dx = 1$$

$$\lambda(100) = 1 \quad \Longrightarrow \quad \lambda = \frac{1}{100}.$$

Therefore,

$$P(50 < X < 150) = \int_{50}^{150} \frac{1}{100} e^{-x/100}\,dx$$

$$= \left[-e^{-x/100}\right]_{50}^{150} = e^{-1/2} - e^{-3/2} \approx 0.3834.$$

Therefore answer is $\boxed{e^{-1/2} - e^{-3/2}}$.

(b) What is the probability that it will function less than 100 hours?

$$P(X < 100) = \int_0^{100} \frac{1}{100} e^{-x/100}\,dx$$

$$= \left[-e^{-x/100}\right]_0^{100} = 1 - e^{-1} \approx 0.6321.$$

Therefore answer is $\boxed{1 - e^{-1}}$.

Teaching Team Notes: It is OK to leave the answer without putting the decimal values. $(1 - 1/e)$ is ok, but leaving the answer as $(-e^{-1} + 1)$ is awkward. The rule of thumb in the exam is try to make the answer easy to read, so a box around the answer is recommended. In homeworks, though, it is upto you, and we may not follow the same in these answers.

3. The joint pdf of X and Y is given by

$$f(x, y) = \frac{6}{7}\left(x^2 + \frac{xy}{2}\right), \qquad 0 < x < 1, \quad 0 < y < 2.$$

(a) Verify that this is indeed a joint density function.
We need to verify that

$$\int_0^1 \int_0^2 f(x, y)\,dy\,dx = 1.$$

Thus,

$$\int_0^1 \int_0^2 \frac{6}{7}\left(x^2 + \frac{xy}{2}\right)\,dy\,dx = \frac{6}{7}\int_0^1 \left(2x^2 + x\right)\,dx$$

$$= \frac{6}{7}\left(\frac{2}{3} + \frac{1}{2}\right) = 1.$$

Since $f(x, y) \geq 0$ on its support, it is a joint density function.

Teaching Team Notes: Note this last line for the exam. You may lose marks if your answer is incomplete.

(b) Compute the density function of X.

The density function of X is

$$\begin{aligned} f_X(x) &= \int_0^2 f(x,y)\,dy \\ &= \frac{6}{7}\int_0^2 \left(x^2 + \frac{xy}{2}\right) dy \\ &= \frac{6}{7}(2x^2 + x), \qquad 0 < x < 1. \end{aligned}$$

Hence,

$$f_X(x) = \begin{cases} \frac{6}{7}(2x^2 + x), & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Teaching Team Notes: Make sure you capture all cases.

(c) $\mathbb{P}\{X > Y\}$ is

The region $X > Y$ is given by $0 < x < 1$ and $0 < y < x$. Therefore,

$$\begin{aligned} \mathbb{P}\{X > Y\} &= \int_0^1 \int_0^x \frac{6}{7}\left(x^2 + \frac{xy}{2}\right) dy\,dx \\ &= \frac{6}{7}\int_0^1 \left(x^3 + \frac{x^3}{4}\right) dx \\ &= \frac{15}{14}\int_0^1 x^3\,dx \\ &= \boxed{\frac{15}{56}}. \end{aligned}$$

4. X and Y are independent continuous random variables. Show that

(a)

$$P\{X + Y \le a\} = \int_{-\infty}^{\infty} F_X(a - y) f_Y(y)\,dy$$

where f_Y is the density function of Y, and F_X is the distribution function of X.

Since X and Y are independent, their joint density is

$$f_{X,Y}(x,y) = f_X(x) f_Y(y).$$

(a) Conditioning on $Y = y$,

$$\begin{aligned} P\{X + Y \le a\} &= \int_{-\infty}^{\infty} P\{X \le a - y \mid Y = y\} f_Y(y)\,dy \\ &= \int_{-\infty}^{\infty} P\{X \le a - y\} f_Y(y)\,dy \\ &= \boxed{\int_{-\infty}^{\infty} F_X(a - y) f_Y(y)\,dy}. \end{aligned}$$

(b)

$$P\{X \le Y\} = \int_{-\infty}^{\infty} F_X(y) f_Y(y)\, dy$$

where f_Y is the density function of Y, and F_X is the distribution function of X.

(b) Similarly, conditioning on $Y = y$,

$$\begin{aligned} P\{X \le Y\} &= \int_{-\infty}^{\infty} P\{X \le y \mid Y = y\} f_Y(y)\, dy \\ &= \int_{-\infty}^{\infty} P\{X \le y\} f_Y(y)\, dy \\ &= \boxed{\int_{-\infty}^{\infty} F_X(y) f_Y(y)\, dy}. \end{aligned}$$

5. In a certain town, there exist 100 buses out of which 1 is red and 99 are blue. XYZ observes a serious accident caused by a bus at night and remembers that the bus was red in color. Hence, the police arrest the driver of the red bus. The driver pleads innocence. Now, a benevolent lawyer decides to defend the distressed bus driver in court. The lawyer ropes in an ophthalmologist to test XYZ's ability to differentiate between the colors red and blue, under illumination conditions similar to those that existed that fateful night. The ophthalmologist's expert deposition to the court is that XYZ sees red objects as red 99% of the time and blue objects as red 2% of the time.

   If you are the defense lawyer how will you argue? That is, compute the probability that the bus was red, when XYZ thought it to be red, possibly correctly, possibly incorrectly. Provide reasons.

   Teaching Team Notes:

   (a) In problems like this, it is good to start with something like: Let X be the event with such-and-such property. We are not sure about the outcome, so we assign a probability to this event. Similarly build other events, or random variables as the case may be, depending on how you like to think about events and random variables.

   (b) In this particular case, consider conditional probability given the wording of the problem statement.

   Solution:

   Let R denote the event that the bus is actually red, and O denote the event that XYZ observes the bus to be red.

   We have

$$P(R) = \frac{1}{100}, \qquad P(R^c) = \frac{99}{100},$$

   and

$$P(O \mid R) = 0.99, \qquad P(O \mid R^c) = 0.02.$$

   By Bayes' theorem,

$$\begin{aligned} P(R \mid O) &= \frac{P(O \mid R)P(R)}{P(O \mid R)P(R) + P(O \mid R^c)P(R^c)} \\ &= \frac{0.99(0.01)}{0.99(0.01) + 0.02(0.99)} \\ &= \frac{0.0099}{0.0297} = \boxed{\frac{1}{3}}. \end{aligned}$$

Thus, the main argument of the defense lawyer is that even after XYZ observed a red bus, the probability that the bus was actually red is only

$$\boxed{\frac{1}{3} \approx 33.3\%}.$$

Hence, there is a substantial probability that the bus was actually blue.

6. $X_1, X_2, \ldots, X_n$ are independent random variables each having the common (canonical) uniform density function

$$f(x) = \begin{cases} 1, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Teaching Team Notes:

(a) In a problem like this which may seem out of reach, first restrict the variables to n=2 and see if you can reason.

(b) When we need the expectation, we need the pdf. For continuous random variables, consider looking at the distribution function and then differentiating it to get the pdf.

(c) Since the problem specification talks of independent, the hint is to think of the joint pdf and then use the product formula.

$E[\text{Max}(X_1, \ldots, X_n)]$ is

Since $X_1, \ldots, X_n$ are independent and uniformly distributed on $(0, 1)$, let

$$M = \text{Max}(X_1, \ldots, X_n),$$

For the maximum,

$$\begin{aligned} F_M(x) &= P(M \leq x) \\ &= P(X_1 \leq x, \ldots, X_n \leq x) \\ &= x^n, \qquad 0 < x < 1. \end{aligned}$$

Thus,

$$f_M(x) = nx^{n-1}.$$

Therefore,

$$E[M] = \int_0^1 x\, nx^{n-1}\, dx = n \int_0^1 x^n\, dx = \boxed{\frac{n}{n+1}}.$$

7. A community consists of 100 married couples. If 50 members of the community die, what is the expected number of marriages that remain intact? Assume that the set of people who die is equally likely to be any of the $\binom{200}{50}$ groups of size 50.

Teaching Team Notes:

(a) In problems like this, you are looking for using the linearity of functions along with indicator variables.

(b) For this problem, this indicator variable is relevant

$$X_i = \begin{cases} 1, & \text{if neither member of couple } i \text{ dies,} \\ 0, & \text{otherwise.} \end{cases}$$

(c) Also pay attention to when the marriage is intact, and when a person dies.

Solution:

Let X be the number of marriages that remain intact. Then

$$X = \sum_{i=1}^{100} X_i.$$

By linearity of expectation,

$$E[X] = \sum_{i=1}^{100} E[X_i] = 100P(X_i = 1).$$

For a particular couple, the probability that neither member dies is

$$P(X_i = 1) = \frac{150}{200} \cdot \frac{149}{199}.$$

Hence,

$$\begin{aligned} E[X] &= 100 \left(\frac{150}{200}\right)\left(\frac{149}{199}\right) \\ &= \frac{11175}{199} \\ &\approx 56.16. \end{aligned}$$